

Reto Final

Leonardo Albarran

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```
library(tidyverse)

## — Attaching core tidyverse packages — tidyverse
## 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.1      ✓ stringr    1.5.1
## ✓ ggplot2     4.0.0      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts —
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
## conflicts to become errors

library(lubridate)
library(readxl)
library(openair)
library(MatchIt)
library(geosphere)
library(janitor)

##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test

library(broom)
library(vars)

## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##   select
##
## Loading required package: strucchange
## Loading required package: zoo
##
```

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## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
##
## Loading required package: sandwich
##
## Attaching package: 'strucchange'
##
## The following object is masked from 'package:stringr':
##
##   boundary
##
## Loading required package: urca
## Loading required package: lmtest

library(lmtest)
library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo

# ==== 1) Parámetros/paths ====
PATH_SRC <- "C:/Users/albar/OneDrive/Escritorio/SIMA/cadereyta_SE3_2024.xlsx"
PATH_NE2 <- "C:/Users/albar/OneDrive/Escritorio/SIMA/apodaca_NE2_2024.xlsx"
PATH_SE2 <- "C:/Users/albar/OneDrive/Escritorio/SIMA/juarez_SE2_2024.xlsx"

# Coordinadas
coords <- tibble::tribble(
  ~Estacion,      ~lat,      ~lon,
  "cadereyta_SE3", 25.60111, -99.99917,
  "apodaca_NE2",  25.77722, -100.18810,
  "juarez_SE2",   25.64583, -100.09530
)

season_of <- function(d) {
  m <- lubridate::month(d)
  dplyr::case_when(
    m %in% c(12,1,2) ~ "winter",
    m %in% c(3,4,5)  ~ "spring",
    m %in% c(6,7,8)  ~ "summer",
    TRUE             ~ "autumn"
  )
}

read_station <- function(path, station_name){
  df <- readxl::read_excel(path, guess_max = 1e5) |>
  janitor::clean_names()
}

```

```

if (!inherits(df$date, c("POSIXct", "POSIXt", "Date"))) {
  if (is.numeric(df$date)) {
    df <- df |> dplyr::mutate(date = as.POSIXct(date * 86400, origin =
"1899-12-30", tz = "America/Monterrey"))
  } else {
    df <- df |>
      dplyr::mutate(date = suppressWarnings(lubridate::ymd_hms(date, tz =
"America/Monterrey"))) |>
      dplyr::mutate(date = ifelse(is.na(date), lubridate::ymd(date, tz =
"America/Monterrey"), date),
        date = as.POSIXct(date, origin = "1970-01-01", tz =
"America/Monterrey"))
  }
} else {
  df <- df |> dplyr::mutate(date = as.POSIXct(date, tz =
"America/Monterrey"))
}

df <- df |> dplyr::mutate(estacion = station_name)

if (!"season" %in% names(df)) {
  df <- df |> dplyr::mutate(season = season_of(date))
}

df |>
  dplyr::select(
    date, co, no, no2, nox, o3, pm10, pm2_5,
    prs, rainf, rh, sr, tout, wsr, wdr, wind_u, wind_v,
    so2, season, photo_index, estacion
  )
}

src <- read_station(PATH_SRC, "cadereyta_SE3")
ne2 <- read_station(PATH_NE2, "apodaca_NE2")
se2 <- read_station(PATH_SE2, "juarez_SE2")

# ==== 4) Utilidades viento/geometría ====
uv_to_wd <- function(u, v){
  (atan2(-u, -v) * 180/pi) %% 360
}

bearing_src_to <- function(src_lat, src_lon, rec_lat, rec_lon){
  as.numeric(geosphere::bearing(c(src_lon, src_lat), c(rec_lon, rec_lat))) %%
360
}

sector_hit <- function(wd, bearing, half_width = 20){
  ang_diff <- ((wd - bearing + 540) %% 360) - 180

```

```

  abs(ang_diff) <= half_width
}

b_ne2 <- with(coords, bearing_src_to(
  lat[Estacion=="cadereyta_SE3"], lon[Estacion=="cadereyta_SE3"],
  lat[Estacion=="apodaca_NE2"], lon[Estacion=="apodaca_NE2"])
)
b_se2 <- with(coords, bearing_src_to(
  lat[Estacion=="cadereyta_SE3"], lon[Estacion=="cadereyta_SE3"],
  lat[Estacion=="juarez_SE2"], lon[Estacion=="juarez_SE2"])
)

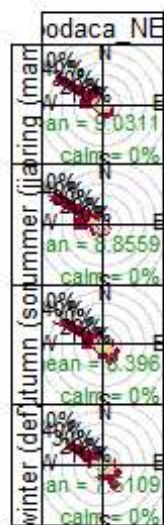
# ==== 5) Preparar pares fuente-receptora ====
prep_pair <- function(source_df, rec_df, bearing_to_rec, ws_min = 2,
half_width = 20){
  df <- source_df |>
    dplyr::select(date, co, no, no2, nox, o3, pm10, pm2_5, so2) |>
    dplyr::rename_with(~ paste0(., "_src"), -date) |>
    dplyr::inner_join(rec_df, by = "date") |>
    dplyr::mutate(
      wd = dplyr::if_else(is.na(wdr) & !is.na(wind_u) & !is.na(wind_v),
        uv_to_wd(wind_u, wind_v), wdr),
      from_refinery = sector_hit(wd, bearing_to_rec, half_width) & wsr >=
ws_min,
      hour = lubridate::hour(date),
      wday = lubridate::wday(date, label = TRUE)
    ) |>
    dplyr::arrange(date)
  df
}

pair_ne2 <- prep_pair(src, ne2, b_ne2)
pair_se2 <- prep_pair(src, se2, b_se2)

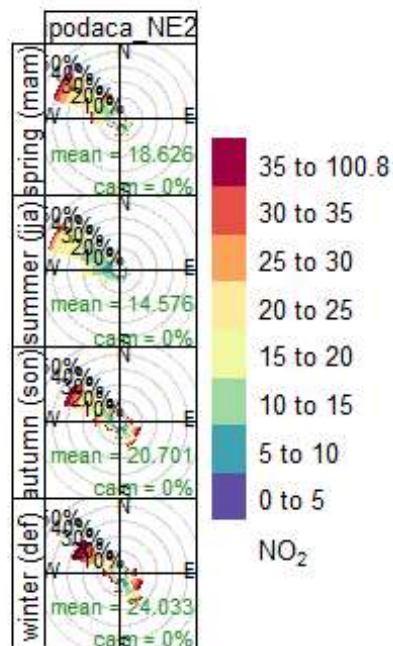
# ==== 6) Diagnósticos openair ====
diag_openair <- function(rec_df, station_label){
  odf <- rec_df |>
    dplyr::transmute(
      date,
      ws = wsr,
      wd = dplyr::if_else(is.na(wdr) & !is.na(wind_u) & !is.na(wind_v),
uv_to_wd(wind_u, wind_v), wdr),
      NO2 = no2, O3 = o3, PM25 = pm2_5,
      season = season, Estacion = estacion
    )
  openair::windRose(odf, ws = "ws", wd = "wd", type = c("Estacion", "season"))
  openair::pollutionRose(odf, pollutant = "NO2", ws = "ws", wd = "wd", type =
c("Estacion", "season"))
  for(pol in c("NO2", "O3", "PM25")){

```

```
      openair::polarPlot(odf, pollutant = pol, ws = "ws", wd = "wd", type =  
"season",  
                        main = paste0("polarPlot - ", station_label, " - ",  
pol))  
    }  
  }  
  
diag_openair(ne2, "Apodaca NE2")
```

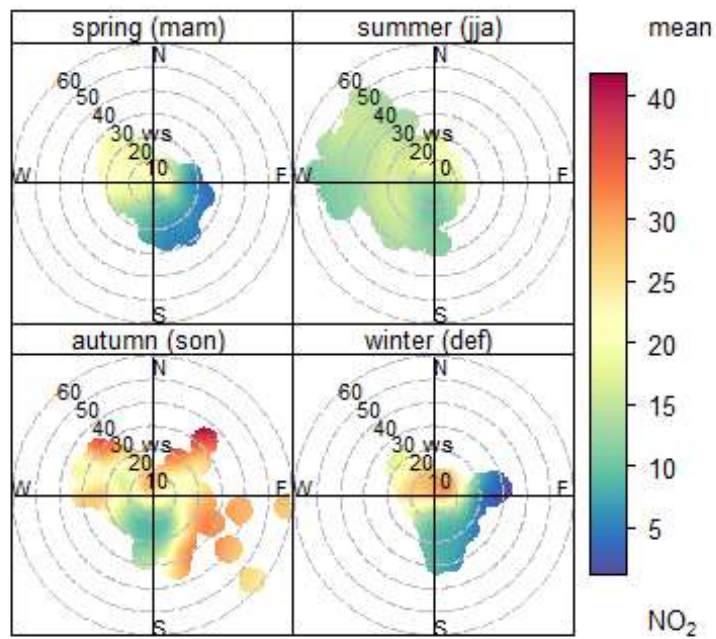


Frequency of counts by wind direction (%)

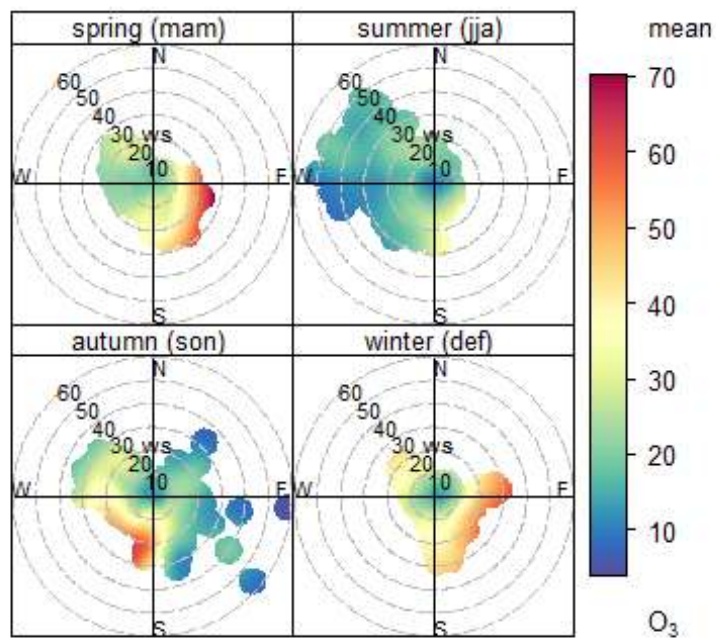


Frequency of counts by wind direction (%)

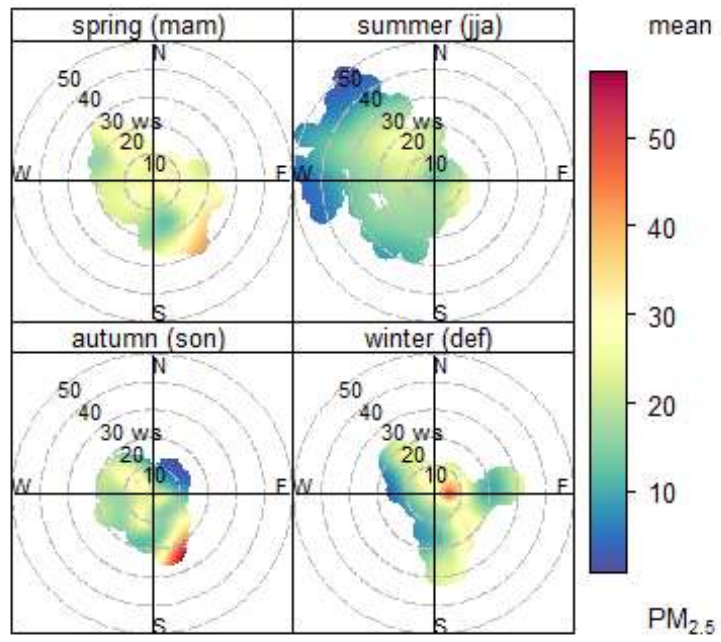
polarPlot - Apodaca NE2 - NO₂



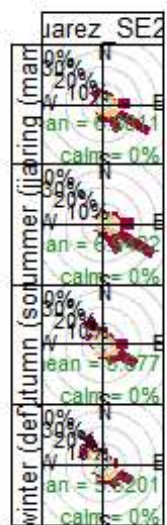
polarPlot - Apodaca NE2 - O₃



polarPlot - Apodaca NE2 - PM_{2.5}

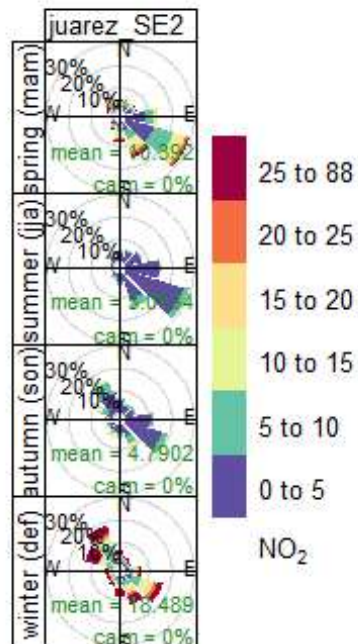


```
diag_openair(se2, "Juárez SE2")
```

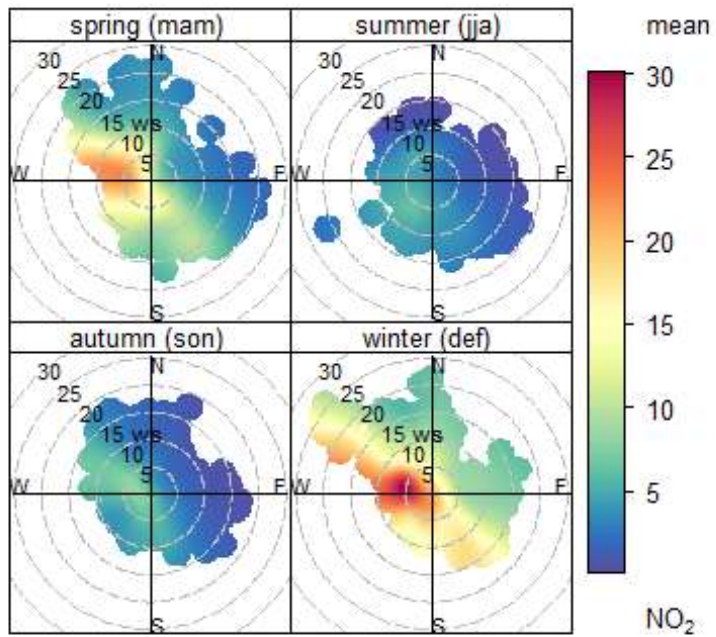
0 to 25.3
(m s⁻¹)

Frequency of counts by wind direction (%)

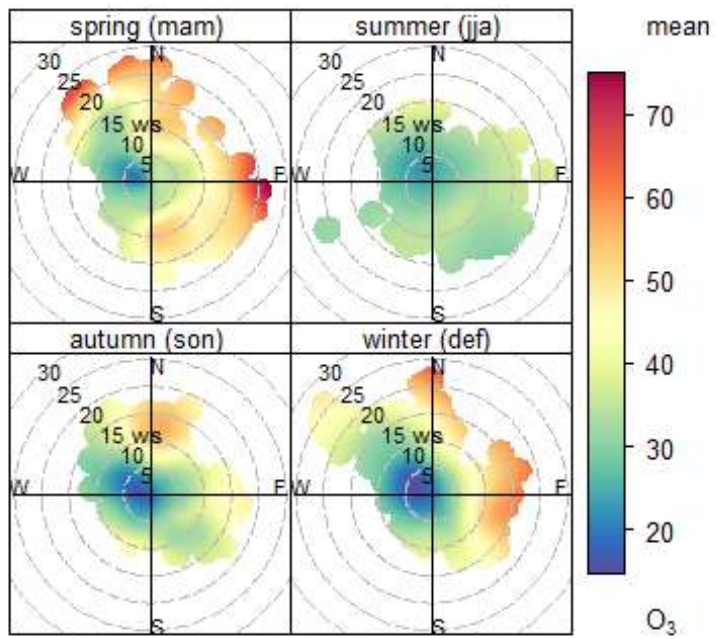


Frequency of counts by wind direction (%)

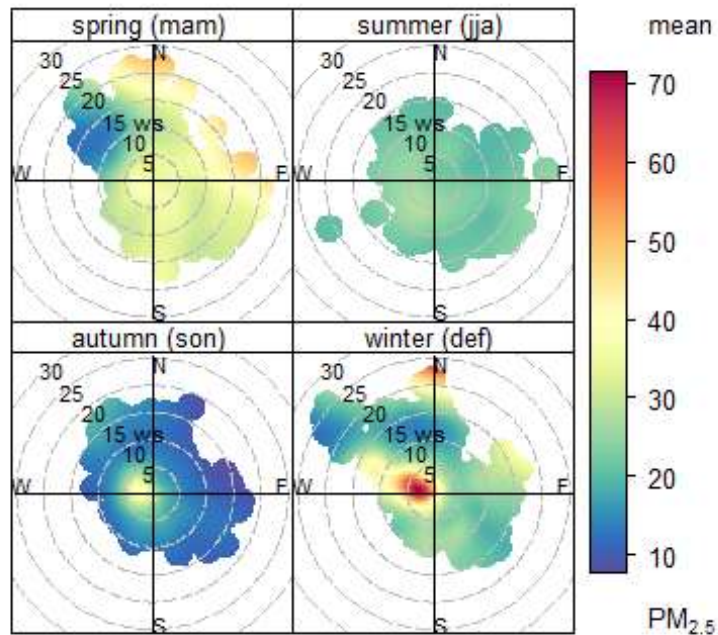
polarPlot - Juárez SE2 - NO₂



polarPlot - Juárez SE2 - O₃



polarPlot - Juárez SE2 - PM_{2.5}



==== 7) Emparejamiento / ATT por temporada ====

Calcula el ATT para un contaminante específico

```
run_matching <- function(df_pair, pollutant = "no2") {
  base <- df_pair |>
    mutate(
      hour = factor(lubridate::hour(date)),
      wday = factor(lubridate::wday(date, label = TRUE))
    ) |>
    dplyr::select(from_refinery, tout, rh, prs, sr, wsr, hour, wday,
!!sym(pollutant)) |>
    tidyr::drop_na()
```

Verifica que haya datos de ambos grupos

```
if (length(unique(base$from_refinery)) < 2L) {
  return(tibble(
    pollutant = pollutant,
    ATT = NA_real_, conf_low = NA_real_, conf_high = NA_real_,
    p_value = NA_real_, n_treated = sum(base$from_refinery),
    n_control = sum(!base$from_refinery),
    note = "Sin variación en from_refinery"
  ))
}
```

Emparejamiento básico con propensity score

```
m <- tryCatch(
```

```

MatchIt::matchit(
  from_refinery ~ tout + rh + prs + sr + wsr + hour + wday,
  data = base,
  method = "nearest", distance = "logit"
),
error = function(e) NULL
)

if (is.null(m)) {
  return(tibble(
    pollutant = pollutant,
    ATT = NA_real_, conf_low = NA_real_, conf_high = NA_real_,
    p_value = NA_real_, n_treated = sum(base$from_refinery),
    n_control = sum(!base$from_refinery),
    note = "Error en MatchIt"
  ))
}

md <- MatchIt::match.data(m)
tt <- t.test(as.formula(paste0(pollutant, " ~ from_refinery")), data = md)

tibble(
  pollutant = pollutant,
  ATT = diff(rev(tt$estimate)), # treated - control
  conf_low = tt$conf.int[1],
  conf_high = tt$conf.int[2],
  p_value = tt$p.value,
  n_treated = sum(md$from_refinery),
  n_control = sum(!md$from_refinery),
  note = "OK"
)
}

# Ejecuta el ATT por temporada y por contaminante
att_by_season <- function(df_pair, label) {
  if (!"season" %in% names(df_pair)) {
    df_pair <- df_pair |> mutate(season = season_of(date))
  }

  pollutants <- c("no2", "so2", "pm2_5", "pm10", "o3")
  results <- list()

  for (pol in pollutants) {
    for (s in unique(df_pair$season)) {
      subset_df <- df_pair |> filter(season == s)
      res <- run_matching(subset_df, pollutant = pol)
      res$season <- s
      res$station_pair <- label
      results[[length(results) + 1]] <- res
    }
  }
}

```

```

    }
  }

  bind_rows(results)
}

# === Ejecuta el cálculo
ATT_NE2 <- att_by_season(pair_ne2, "Cadereyta->NE2")

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

ATT_SE2 <- att_by_season(pair_se2, "Cadereyta->SE2")

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

ATT <- bind_rows(ATT_NE2, ATT_SE2) |>
  arrange(station_pair, pollutant, season)

print(ATT)

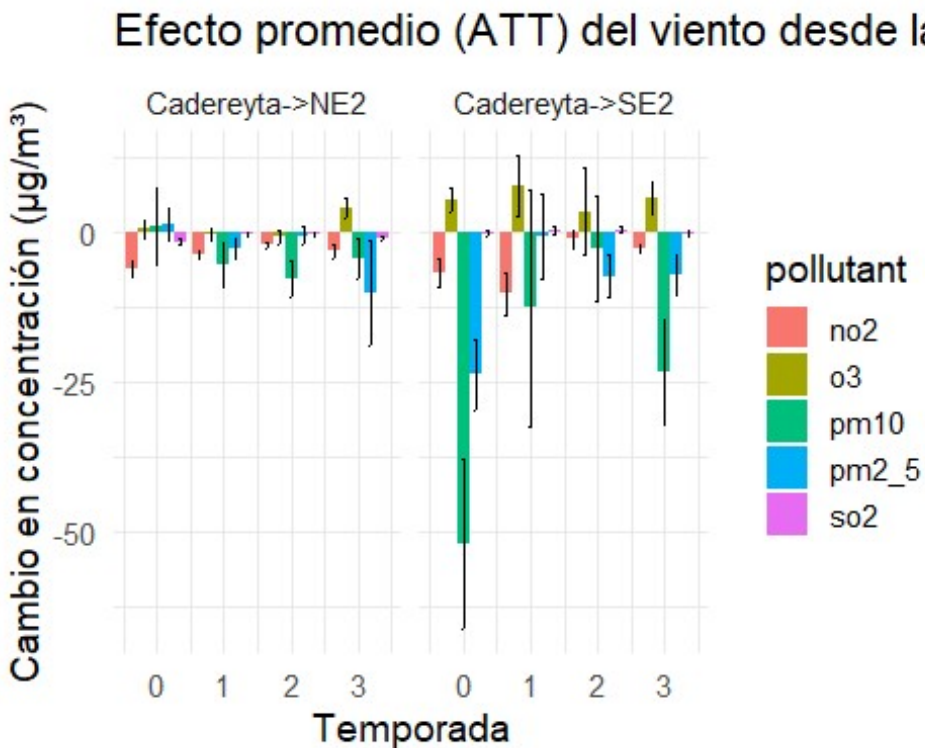
## # A tibble: 40 × 10
##   pollutant    ATT conf_low conf_high p_value n_treated n_control note
##   <chr>      <dbl>   <dbl>   <dbl>   <dbl>   <int>   <int> <chr>
##   <dbl>
## 1 no2      -6.22    -7.56    -4.87  4.32e-19    656     656 OK
## 2 no2      -3.67    -4.36    -2.99  3.95e-25    900     900 OK
## 3 no2      -2.09    -2.66    -1.53  6.78e-13    766     766 OK
## 4 no2      -3.19    -4.29    -2.08  2.03e- 8    682     682 OK
## 5 o3        0.538   -1.08     2.15  5.13e- 1    656     656 OK
## 6 o3      -0.429   -1.49     0.629 4.27e- 1    900     900 OK
## 7 o3      -0.826   -1.96     0.309 1.54e- 1    761     761 OK
## 8 o3        4.03     2.41     5.64  1.08e- 6    682     682 OK
## 9 pm10     0.926   -5.46     7.32  7.76e- 1    655     655 OK
## 10 pm10    -5.39    -9.14    -1.64  4.83e- 3    900     900 OK

```

```
## # i 30 more rows
## # i 1 more variable: station_pair <chr>

library(ggplot2)

ggplot(ATT, aes(x = season, y = ATT, fill = pollutant)) +
  geom_col(position = position_dodge()) +
  geom_errorbar(aes(ymin = conf_low, ymax = conf_high),
               width = 0.2, position = position_dodge(0.9)) +
  facet_wrap(~ station_pair) +
  labs(title = "Efecto promedio (ATT) del viento desde la refinería",
       y = "Cambio en concentración (µg/m³)",
       x = "Temporada") +
  theme_minimal(base_size = 14)
```



```
#write.xlsx(ATT, "ATT.xlsx")
```